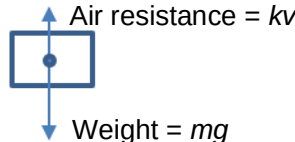


Q.	Solution															
1(a)	 There are two opposing forces acting on the object as it falls – its own weight and air resistance. As the object falls, air resistance increases as the object's velocity increases, causing the resultant force on object to decrease. By Newton's second law, the acceleration of the object likewise decreases. Eventually, when the object reaches a velocity such that the air resistance is equal in magnitude to its weight, the resultant force and acceleration of the object become zero and the object travels at constant velocity according to Newton's first law of motion.															
1(b)(i)	 According to Newton's second law of motion: resultant force on object = $mg - kv = ma$ $\Rightarrow mg - ma = kv$ $\Rightarrow (g - a) = kv/m$															
1(b)(ii)	(The gradient of the tangent to the velocity-time graph in Fig. 1.1 at various points gives the instantaneous acceleration at those points.) For eg. The gradient of the velocity-time graph at $v = 30 \text{ m s}^{-1} = (44 - 10)/6.0 = 5.7 \text{ ms}^{-2}$ The completed table is shown below: <table><tr><th>velocity v/ms^{-1}</th><th>acceleration a/ms^{-2}</th><th>$(g - a)/\text{ms}^{-2}$</th></tr><tr><td>0</td><td>9.8</td><td>0.0</td></tr><tr><td>20</td><td>8.2</td><td>1.6</td></tr><tr><td>30</td><td>5.7</td><td>4.1</td></tr><tr><td>40</td><td>0.0</td><td>9.8</td></tr></table>	velocity v/ms^{-1}	acceleration a/ms^{-2}	$(g - a)/\text{ms}^{-2}$	0	9.8	0.0	20	8.2	1.6	30	5.7	4.1	40	0.0	9.8
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40	0.0	9.8														
1(b)(iii)	Since $(g - a) = kv/m$, if the student's suggestions are correct, the value of the ratio k/m for $v = 20 \text{ m s}^{-1}$ and 30 m s^{-1} should agree with that obtained at $v = 40 \text{ m s}^{-1}$, which is calculated to be: $(g - a)/v = 9.8/40 = 0.25 \text{ s}^{-1}$.															

	At speed $v = 20 \text{ m s}^{-1}$, $(g - a)/v = 1.6/20 = 0.08 \text{ s}^{-1}$. At speed $v = 30 \text{ m s}^{-1}$, $(g - a)/v = 4.1/30 = 0.14 \text{ s}^{-1}$. Since the ratio k/m is not a constant, the student's suggestion is incorrect.
2(a)(i)	Both trolleys could not be stationary at the same time as this would imply that the